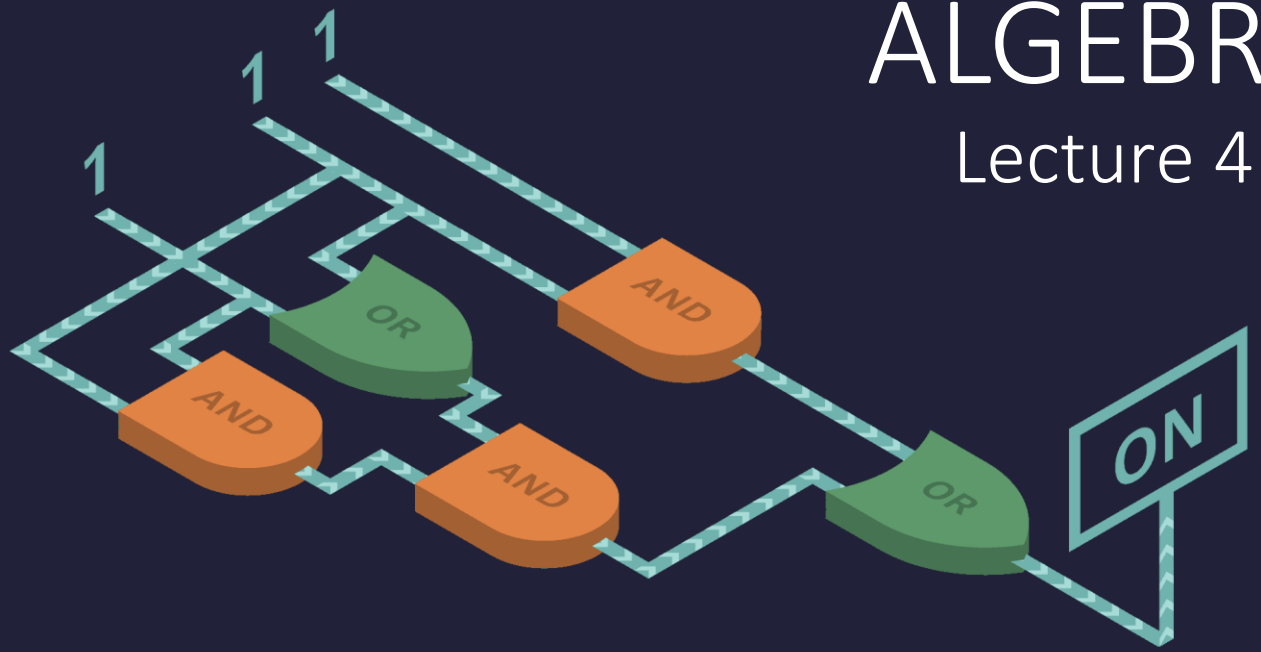
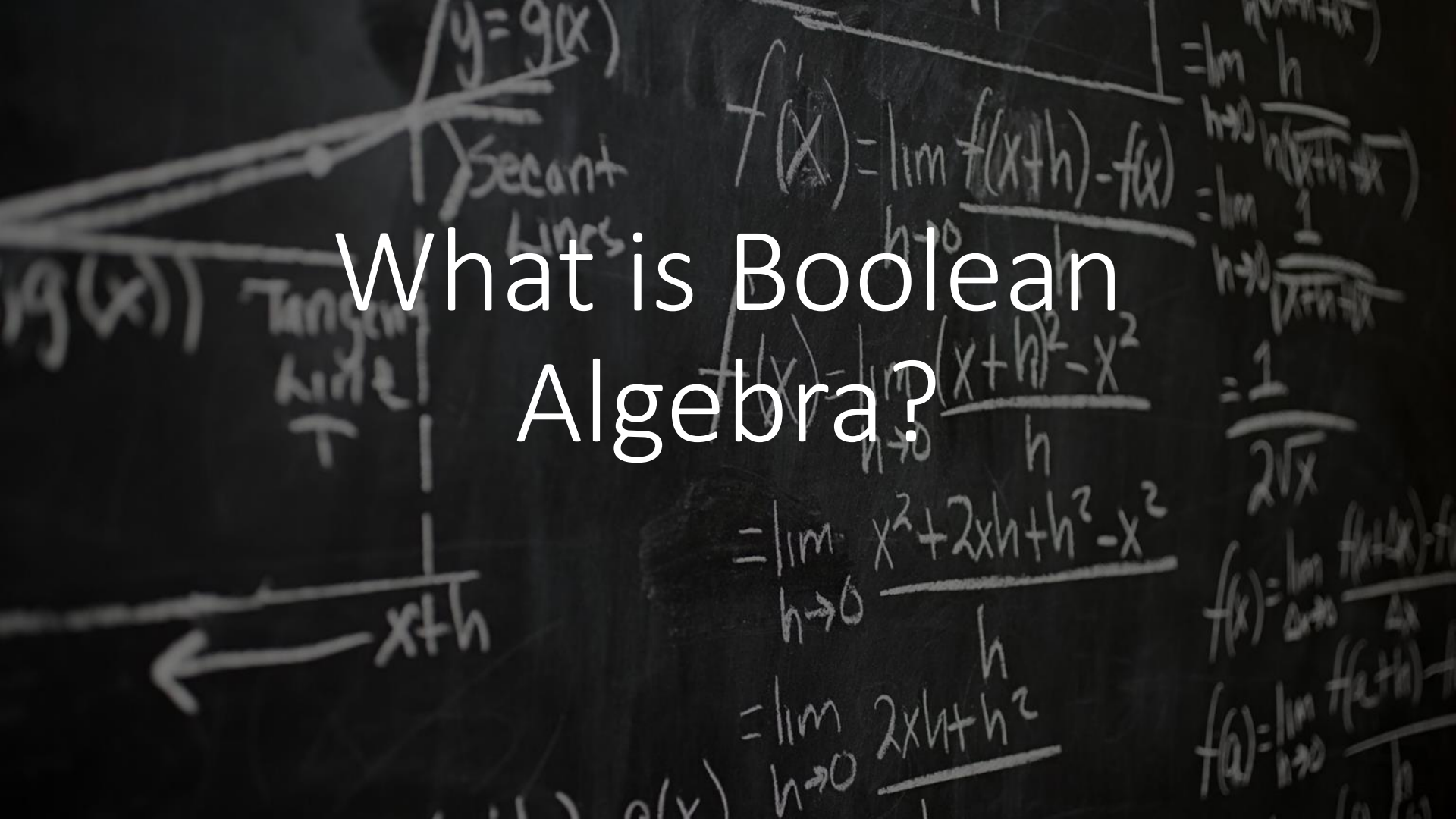
BOOLEAN ALGEBRA

Lecture 4




What is Boolean Algebra?

Boolean Algebra



Boolean Algebra is used to analyze and simplify the digital (logic) circuits.



It uses only the binary numbers (0 and 1).



It is also called as Binary Algebra or Logical Algebra.

Basic Boolean Expressions

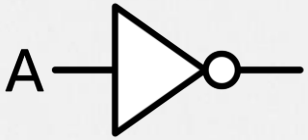
Basic Boolean Expressions

- One or several Boolean Variables combined with logical operations (gates) gives us Boolean expressions.

- Each logic gate is associated with its own Boolean expressions.

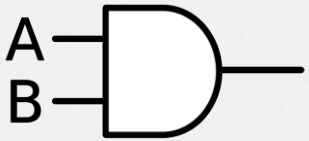
Three basic logic gates

1. NOT



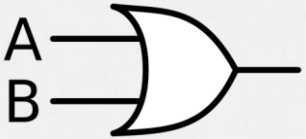
$Y = \overline{A}$

2. AND



$Y = A \cdot B$

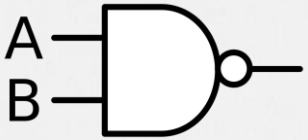
3. OR



$Y = A + B$

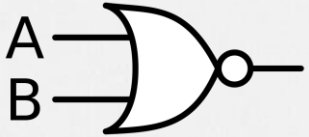
Derived logic gates

1. NAND



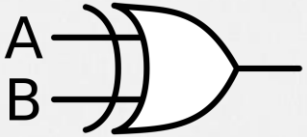
$Y = \overline{A \cdot B}$

2. NOR



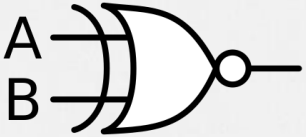
$Y = \overline{A + B}$

3. XOR



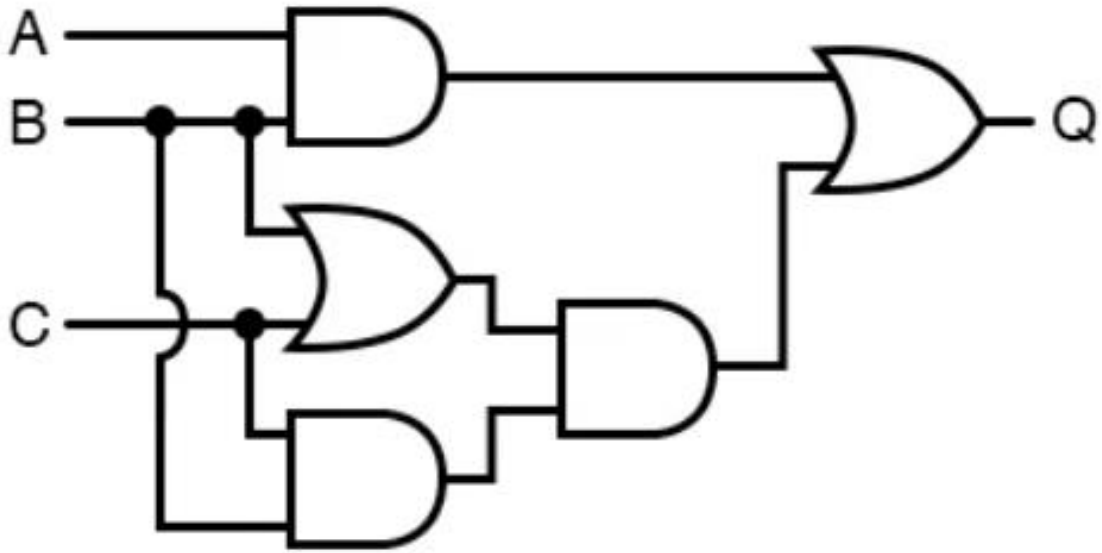
$Y = A \oplus B$

4. XNOR

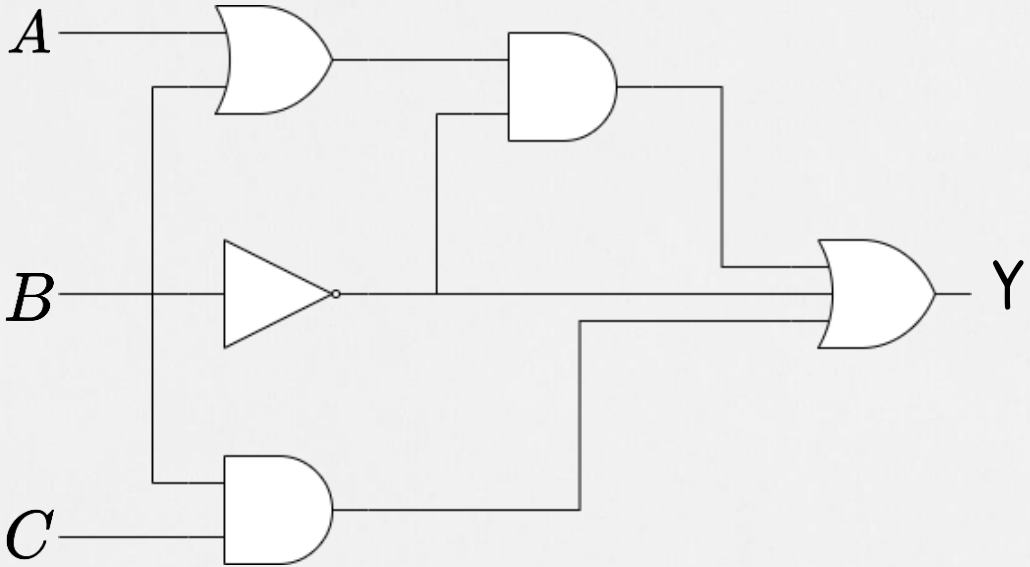


$Y = \overline{A \oplus B}$

Obtain the logic expression for the following circuit.



Your turn!



Draw a logic circuit for the following Boolean expression.

$$Y = AB + (B+C) + C'$$

Boolean Algebra Laws

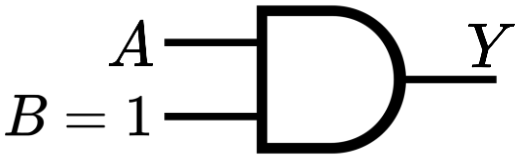
Boolean Algebra Laws

The rules that need to be followed to simplify the complex logic expressions.



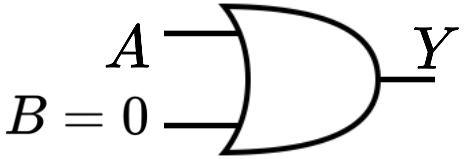
1. Identity Law

$$A \cdot 1 = A$$



A	B	Y
0	1	0
1	1	1

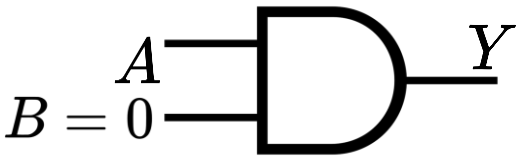
$$A + 0 = A$$



A	B	Y
0	0	0
1	0	1

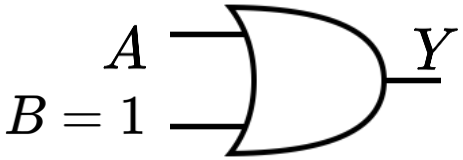
2. Null Law

$A.0 = 0$ 



A	B	Y
0	0	0
1	0	0

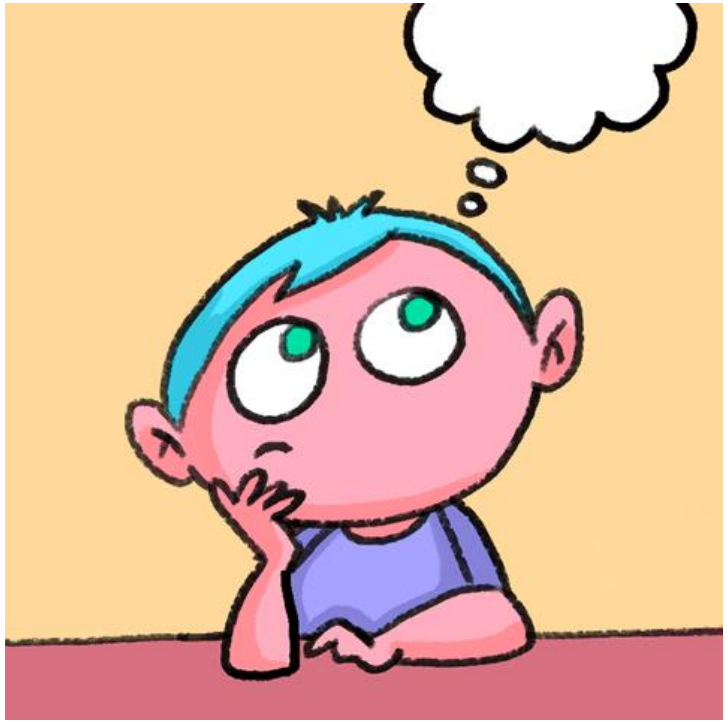
$A + 1 = 1$ 



A	B	Y
0	1	1
1	1	1

Can you guess?

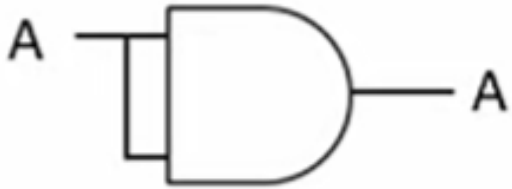
- $A \cdot A =$
- $A + A =$

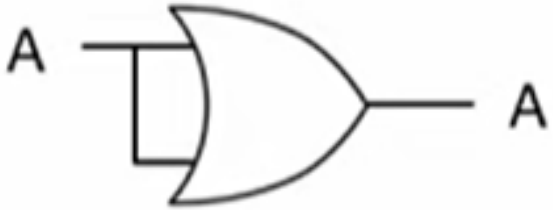


3. Idempotent Law

$$A \cdot A = A$$

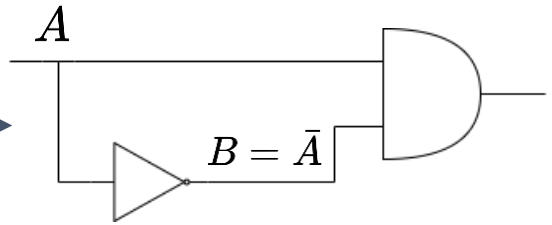
$$A + A = A$$





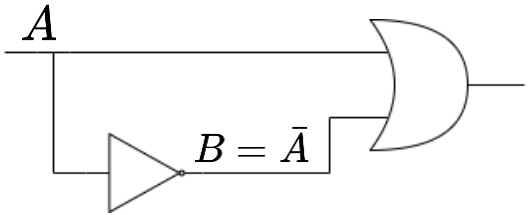
4. Inverse/Complement Law

$$A.\bar{A} = 0$$



A	B	Y
0	1	0
1	0	0

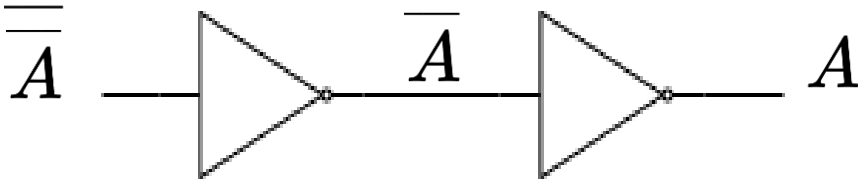
$$A + \bar{A} = 1$$



A	B	Y
0	1	1
1	0	1

5. Double Negation Law

$$\overline{\overline{A}} = A$$



6. Commutative Law

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

7. Associative Law

$$A + (B + C) = (A + B) + C$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

8. Distributive Law

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

9. Absorption Law

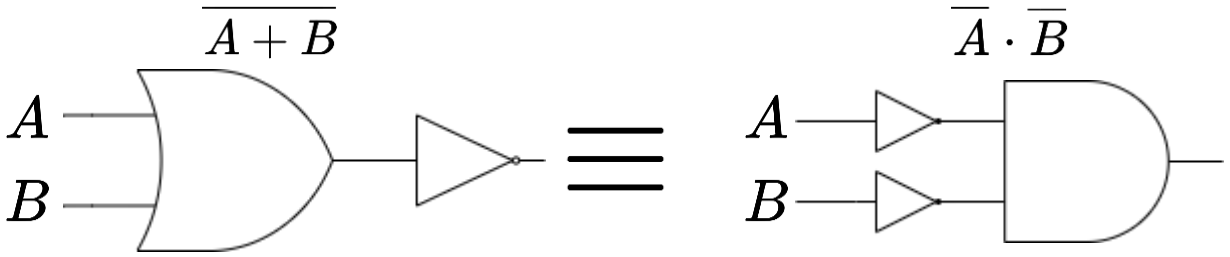
$$A(A+B) = A$$

$$A + AB = A$$

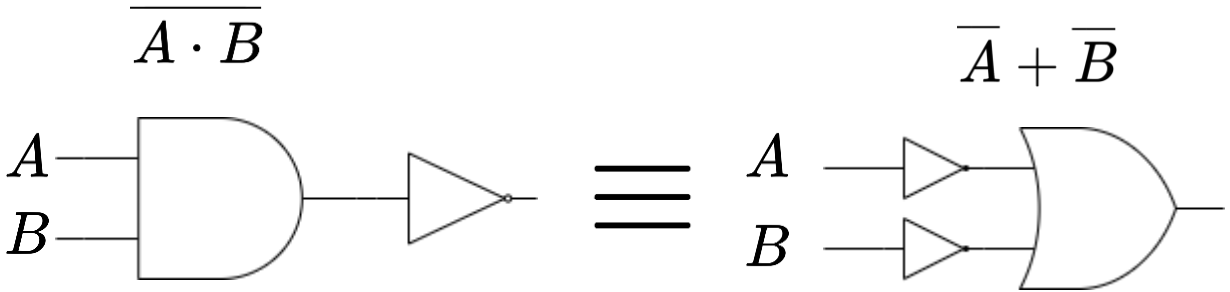
10. DeMorgan's Law

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$



A NOR gate is equivalent to an inversion followed by an AND

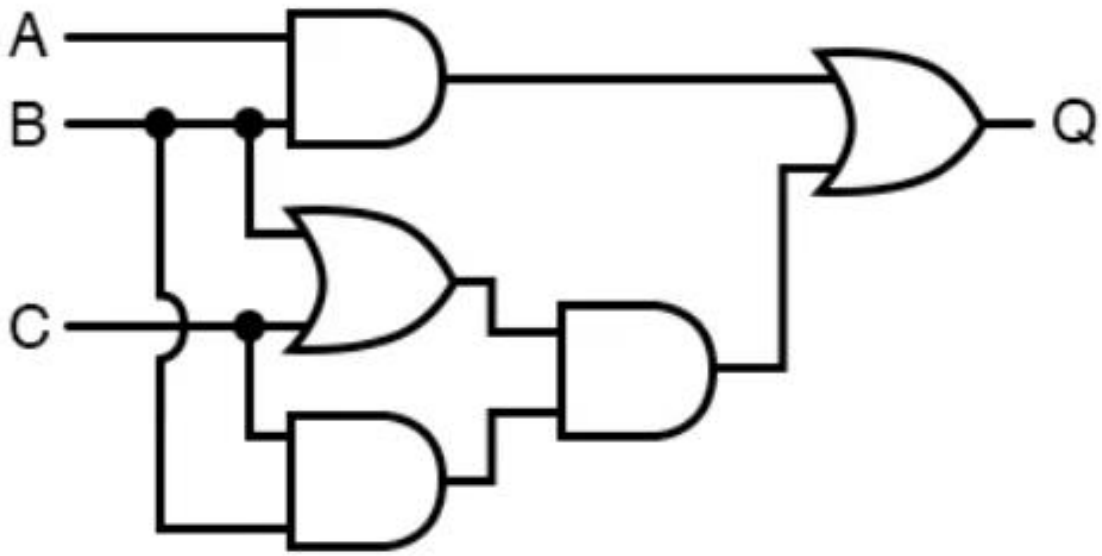


A NAND gate is equivalent to an inversion followed by an OR

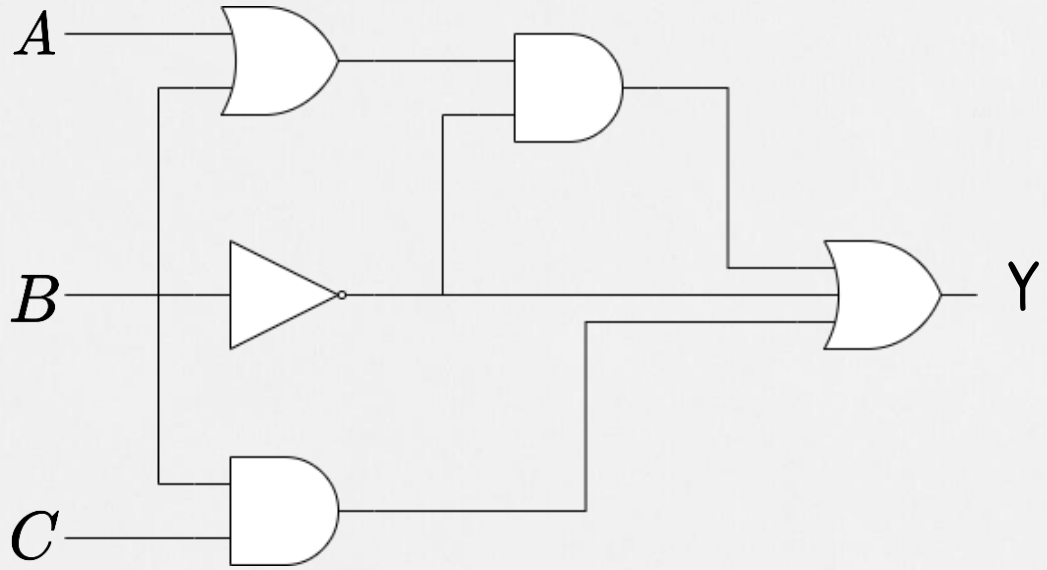
Name	AND form	OR form
Identity law	$1A = A$	$0 + A = A$
Null law	$0A = 0$	$1 + A = 1$
Idempotent law	$AA = A$	$A + A = A$
Inverse law	$A\bar{A} = 0$	$A + \bar{A} = 1$
Commutative law	$AB = BA$	$A + B = B + A$
Associative law	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
Distributive law	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption law	$A(A + B) = A$	$A + AB = A$
De Morgan's law	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$

Double Negation Law => A'' = A

Obtain the logic expression for the following circuit and simply it using the Boolean algebra laws



Obtain the logic expression for the following circuit and simply it using the Boolean algebra laws



Thanks!

Any questions?